PMTRD: from platelet RNA to biomarker reliability

Discovery data

GSE89843

633 participants

28,089 genes

402 NSCLC
231 healthy

Site + RNA QC

Cohort metadata

1 Nested evaluation

Repeat over 5 outer folds

Training fold → **Re-select genes** Fold-specific panel → **Fit XGBoost** → **Predict held-out fold** Out-of-fold predictions

Test fold → Out-of-fold predictions

Held-out patients bypass selection and fitting

2 Locked external validation

Full-cohort selection 27-gene panel → **Fit + freeze XGBoost** Discovery $\log_2$-CPM → Locked → **External predictions** No retraining / tuning

Selection details: Figure S8

GSE68086 · n = 94

3 Audit panels, predictions and cohort structure

Predictive audit $\mathcal{R}_P$

Predictions + labels

↓ **AUC + calibration**

Nested + external tests

Molecular audit $\mathcal{R}_M$

Resample + split halves

↓ **Stability · ψ · replication**

Gene / pathway replication

Technical audit $\mathcal{R}_T$

Site / contamination / alignment

↓ **Re-evaluate transfer**

AUC shift + gene coverage

PMTRD reliability card: report each dimension separately with 95% intervals